# Mike Dean's Railroad Book

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# Preface

# Part I

# Club Module

# Chapter 1

# Wasatch N Scale Module

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## 1.1 Wasatch N Scale Club

I joined the Salt Lake City based railroad club that later became the NRMA Northern Division, and became aware that there was a club devoted entirely to N scale modeling. In the spring of 2013 I contacted their president, Bob Gerald, and arranged to come to the Evanston train show that was being held in July. I was invited to bring some trains, as the club would be set up and they needed people to run train. I didn't really understand that, but my wife and I drove up for an afternoon and ran trains around the club layout.

What I really did not understand was that they really *needed* interested people to run trains. Having subsequently attended numerous train shows, including several in Evanston, it becomes important to make sure that trains are running throughout the show, because the public (especially children) simply love to watch trains go.

At my first club meeting, I heard about the club standards, and learned a fair amount about N-trak (upon which our clubs standards are based). It was pointed out that full membership required building a module, since full members could vote on changing module standards and it was considered unreasonable to vote on such matters unless you built a

module yourself. Modules range in size, but I was informed that the club really needed some more basic two by four foot modules, rather than additional large or corner modules. Thus, I embarked on the adventure of building my first module.

## 1.2 Initial Module Planning

## 1.3 Building a Mountain in the Kitchen

## 1.4 Wiring and Tortoises

## 1.5 Adding Signaling to the Module

## 1.6 Fixing a Disaster

## 1.7 Status 2019 - 2020

## 1.8 Conclusions about Modules and Clubs

Modules are an excellent way for a beginning or advanced modeler to get something meaningful off the ground. When I started my module, I had only a rudimentary oval layout in my basement, and it was simply not tenable to learn about all the various technologies quickly in the context of a basement layout. The club activities at train shows provided me with a timeline, including deadlines, and this led to sporadic but regular progress on the module. I have also used the module as an experimental platform for different scenery techniques, such as static grass.

I cannot say enough how important a club can be to a beginner model railroader. The direct hands on assistance I received by Robert Arneson and Bob Gerald was critical - this enabled me to have my module ready for the first Hostler show, complete with Tortoise controlled turnouts and rudimentary scenery. In one afternoon, they propelled me through what would have taken months without their assistance.

# Part II

# Initial 4 by 8 Basement Design

# Chapter 2

# Test Chapter

# Part III

# Current Basement Designs

# Chapter 3

# Test Chapter

# Part IV

# JMRI Configuration Using GIT

# Chapter 4

# Test Chapter

# Part V

# 3D Printing

# Chapter 5

# 3D Printing for Railroad

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## 5.1 Ender 3 Printer from Creality

I have mostly ignored 3d printing because of the high cost of these printers. In December 2018, about two weeks before Christmas with all kinds of family in the house, I ordered the Ender 3 printer from Creality. This printer had (and still has) established notoriety because of its relatively high quality print capability and a cost that was sometimes under $200. The printer comes as a kit that has to be assembled, but the major complex pieces are already assembled. I found an incredibly helpful video on YouTube that helped understand the instructions. The major issue is assuring that the frame is put together square.

After assembly, I ran a test print from the included SD card, ending up with a dog. After this, of course, one has to find things to print or figure out how to design items on the computer using various pieces of software. ThingiVerse is a web site with thousands of files that can be downloaded and printed, but certain things simply cannot be printed with a fused deposition modeling (FDM) type of printer. The Ender 3 is an FDM printer. This heats up plastic and deposits it in layers, and after depositing one layer, the print head rises 0.1 to 0.2 mm (depending on settings) and deposits the next layer. Extremely small or detailed parts do not print well, and it is possible to end up with a mess of plastic spaghetti with certain patterns. If I want to get into small detailed parts, I will be forced to buy a different type of 3d printer based on stereolithography (resin printers). At the time of this writing, in December 2019, I will defer on this next technology.

## 5.2 Original Prusa i3 MK3S Printer

In July 2020, in the midst of my fifth month in pandemic experience, I ordered the Prusa printer in kit form, which arrived in August. The kit was challenging to build, but now I understand how the printer works. I was reasonably careful, and the machine has worked very well. Its primary advantages include filament detection, autoloading, autounloading, automatic bed leveling, and Z-level adjustments. It is also a direct extrusion machine (not a Bowden tube machine) and has a far superior extrusion mechanism compared with the Ender.

## 5.3 Setting up Octoprint with Raspberry Pi

I decided to try a piece of software called Octoprint. The first step was to 3d print some parts that will hold a camera for the Raspberry PI, as well as some enhancements of the Ender 3. This was bizarre (enhancing my printer by printing enhancement parts). After I replaced the Ender 3 with my Prusa machine, I had to print a different set of parts, which was easily accomplished. The camera enables me to see the status of the print, and also allows creating a time lapse movie of the overall print.

The settings are to make sure my desktop computer is on the DeanEERO, because the Pi is on DeanEERO. Then I simply go to a browser and aim at http://octopi.local and it logs into the Raspberry Pi with the interface for controlling the printer. I don't have to mess with the operating system or finding the software, as this is automatically set up. I need to upload files to the Pi in order for them to get printed, or alternatively, I can print from the SD card that is in the printer. I have noticed that if I print from the SD card, Octoprint does not have access to the GCODE, time lapse does not work, and it is possible to accidentally abort the print. Therefore, I generally start an SD card print from the printer LCD and do not make any adjustments via the OctoPrint interface.

The user name is mdean77 and the password is Usual98a.

## 5.4 Development Process for 3d Parts

There are several types of software available for doing 3D design on the computer. Some are free, some are not. I quickly became aware of Fusion 360, which is a high end product that turns out to be completely free for hobbyists. It has a steep learning curve, which I have attempted to climb, and it is certainly not the easiest option available. There are other products but Fusion is where I landed. After completing a design, I export the file as an STL file.

The STL file now needs to be "sliced" into the individual layers that the FDM printer will lay down, one by one. The slicer software produces GCODE, which provides the printer with exact instructions for conducting the print. I have three slicer software packages. Cura was

the first I used because it was included with the Ender 3, was free, and is very functional. I purchased Simplify3D after a year, and this software allows having multiple processes. This means that I can change settings in the middle of a print. Finally, I have been using PrusaSlicer, which is provided in conjunction with the Original Prusa i3 MK3S printer that has replaced my Ender 3. Each of these slicers has advantages and disadvantages, and it is unclear that any one of them is particularly superior to the others.

## 5.5 Using Parts to Hide Accessory Decoders

To learn Fusion 360 I found a tutorial that built a box with a snap on lid. This led me to the idea of creating small boxes that could contain the accessory decoders used to control my turnouts (Figure 5.1). I have a large number of turnouts and it is obviously not an attractive seen to have decoders attached in haphazard manner. For this reason, I created a box design that would contain the decoder and simplify the overall appearance (Figure 5.2). Finally, I could make different versions depending on how many decoders were located in different areas of the layout, as shown in Figure 5.3.

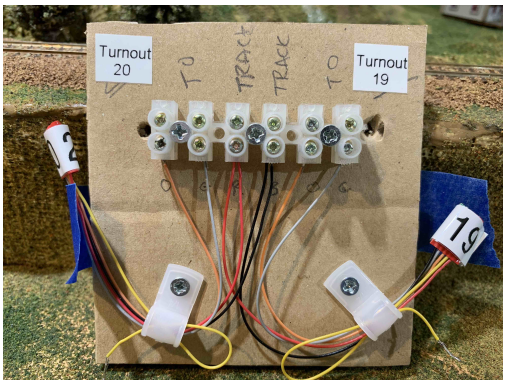



Figure 5.1: Accessory decoders taped to underside of layout.

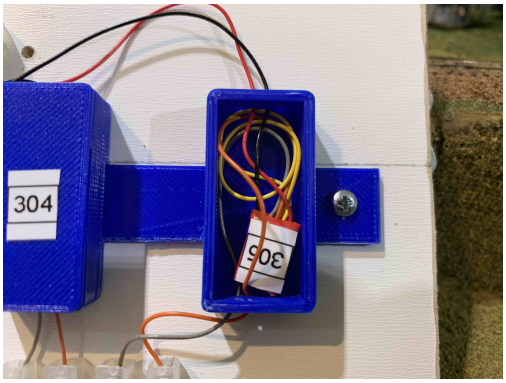
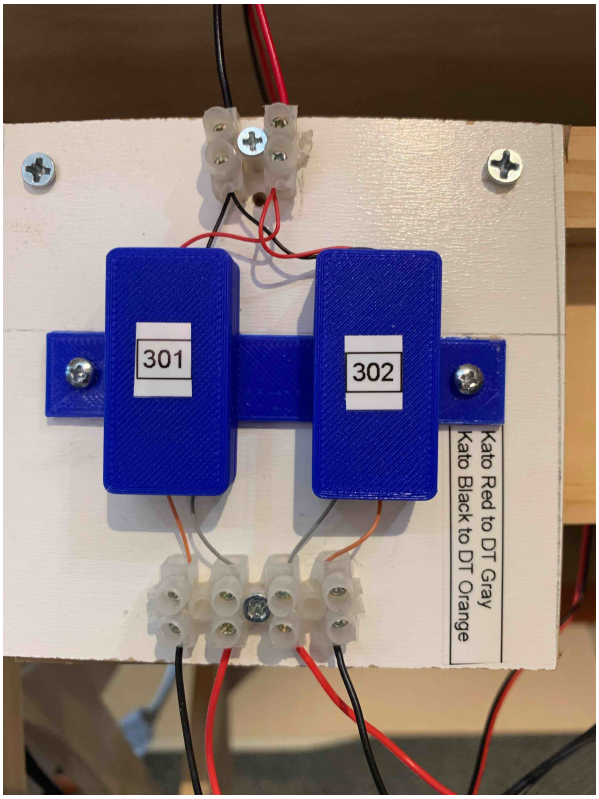
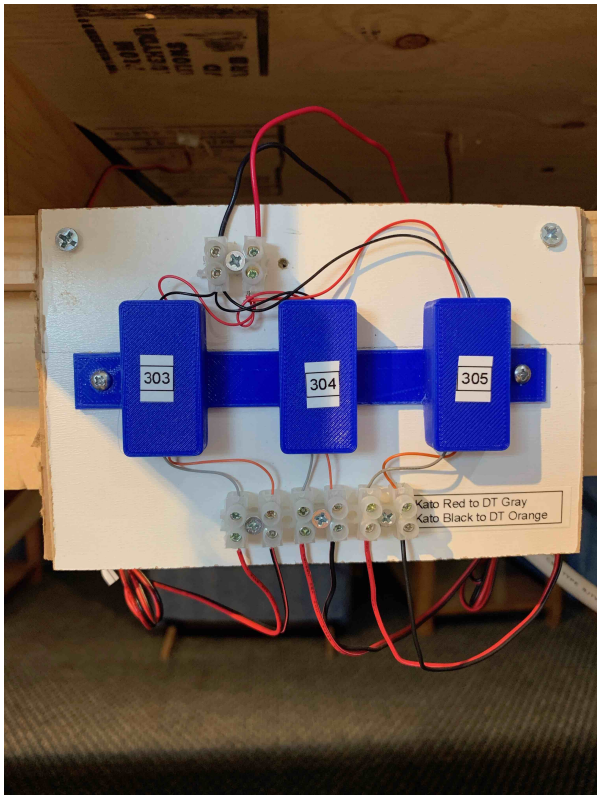


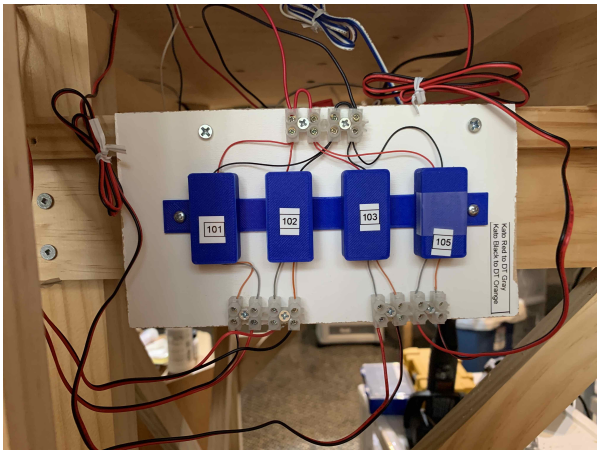

Figure 5.2: Decoder in 3D box, simplifying the appearance.



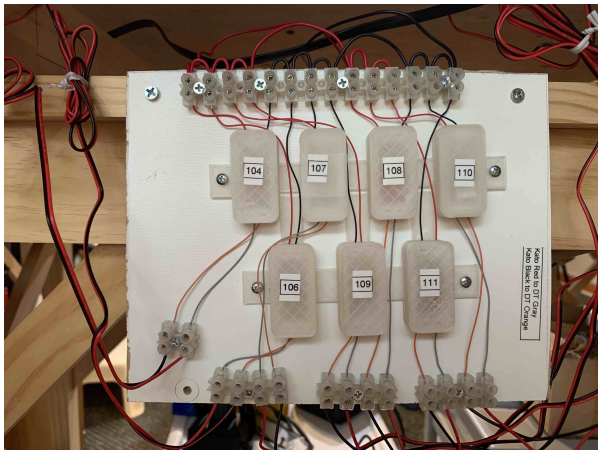

(a) Print of two boxes




(b) Print of three boxes




(c) Print of four boxes




(d) Print of seven boxes

Figure 5.3: 3D assorted size printed boxes to contain accessory decoders for Kato turnouts.

## 5.6 Printing Structures and Rolling Stock

To be written.

# Part VI

# Test Track

# Chapter 6

# Test Chapter

# Part VII

# Electronics

# Chapter 7

# Test Chapter